

WASP-52b

R-0519

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180724 · created 2026-07-20T23:28:03+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458323.8727 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.035
Transit depth (Rp/Rs)^2	4.47 +/- 1.48 [%]
Orbital Inclination (inc)	82.87 +/- 1.52
Airmass coefficient 1 (a1)	1.043 +/- 0.021
Airmass coefficient 2 (a2)	-0.036 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.97 %
Best Comparison Star	#2 - [298, 220]
Optimal Method	PSF photometry
Transit Duration (day)	0.055 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.211 $\pm$ 0.035 vs 0.1646 (deviation 1.33 σ)
Duration fit vs published	0.055 d vs 0.0758 d (deviation 0.95 σ)
Mid-time O-C	10.7 min
Depth SNR	6.0
Residual scatter	1.97 %

Light curve

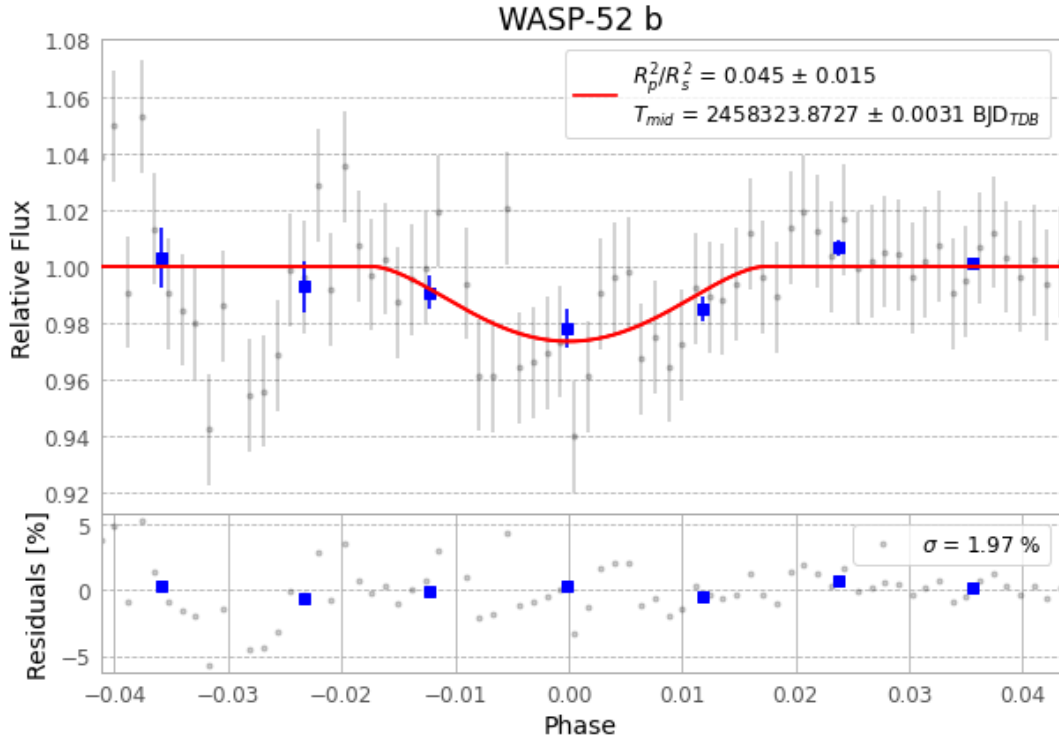


Figure 1 — FinalLightCurve_WASP-52 b_2018-07-23.png (SHA-256 48edbc3241d6f4d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2baadd8f8db05faf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

75 FITS frames (49 MB), WASP-52180724065712.FITS → WASP-52180724103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180724.log (1fdeb5a5301a...), FinalLightCurve_WASP-52 b_2018-07-23.png (48edbc3241d6...), FinalLightCurve_WASP-52 b_2018-07-23.csv (946e86f34613...)

Compute resources

Wall time 210.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)